

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

M.Tech Degree S2 (S,FE) / S2 (WP) (S,FE) Examination December 2024 (2022 Scheme)

Course Code & Name: 222TCS100 ADVANCED DATA STRUCTURES AND ALGORITHMS

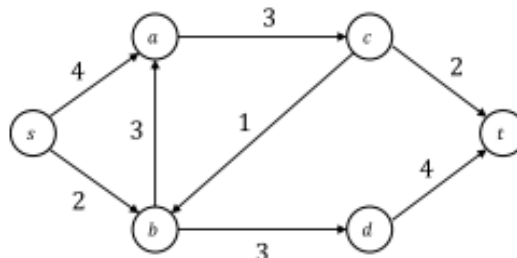
Max. Marks: 60

Duration: 2.5 Hours

PART A**Answer all questions. Each question carries 5 marks**

Marks

- 1 Derive the asymptotic bound for the recurrences.
 - (a) $T(n) = 2T(n/2) + n / \log^2 n$ (2.5)
 - (b) $T(n) = 9T(n/3) + n^2$ (2.5)
- 2 What is a binomial tree? Enumerate its properties. (5)
- 3 Apply Ford Fulkerson algorithm on the following network to obtain the maximum flow across the network. (5)

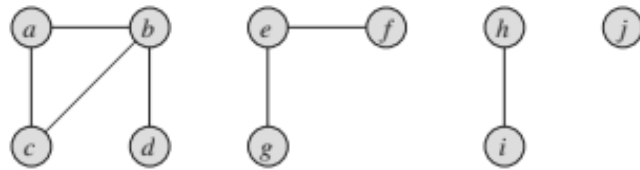


- 4 Apply Miller Rabin algorithm to test whether the number 561 is prime or not. (5)
- 5 Design an approximation algorithm for the traveling-salesman problem to find the minimum cost tour. Derive the approximation ratio $\rho(n)$, of the algorithm. (5)

PART B**Answer any 5 questions. Each question carries 7 marks**

- 6 (a) Apply RABIN-KARP algorithm to search for the pattern $p = \text{"abra"}$ in text $T = \text{"abracadabra"}$. where $q=101$. (4)
- (b) What is the total cost of executing n of the stack operations PUSH, POP, and MULTIPOP, assuming that, the stack begins with s_0 objects and finishes with s_n objects? (3)
- 7 (a) Construct a Binomial heap using the numbers -1, 5, 7, 8, 2, 10, 100, 1. (4)

- (b) Illustrate how disjoint-set data structure computes the connected components of the undirected graph shown below: (3)



- 8 (a) Explain the significance of child cut member in Fibonacci heap with example. (4)
- (b) Demonstrate simple union operation using linked list set representation. (3)
- 9 (a) Find the maximum bipartite matching of the graph $G(L, R, E)$, where $L: \{a, b, c, d, e\}$, $R: \{1, 2, 3, 4, 5\}$ and $E = \{(a, 1), (a, 2), (b, 1), (c, 2), (c, 3), (d, 3), (d, 4), (d, 5), (e, 5)\}$. (5)
- (b) State the max-flow min-cut theorem. (2)
- 10 (a) State and prove Euler's theorem. (3)
- (b) With suitable examples differentiate Monte Carlo and Las Vegas algorithms. (4)
- 11 (a) State and prove Fermat's theorem. (3)
- (b) With suitable example explain the probabilistic algorithm for verifying matrix multiplication problem. (4)
- 12 (a) Illustrate with example that APPROX-VERTEX-COVER is a polynomial-time 2-approximation algorithm. (4)
- (b) Explain the approximation algorithm for traveling salesperson problem. (3)
